Open-Source Physical Design Flows for Monolithic and Heterogeneous 3D ICs using OpenROAD

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***Abstract*—This tutorial presents a practical introduction to open-source physical design methodologies for monolithic and heterogeneous three-dimensional integrated circuits (3D ICs) using the OpenROAD framework. As conventional CMOS scaling approaches its limits, 3D integration has emerged as a promising solution for improving performance, energy efficiency, and integration density. Despite growing industrial interest, most publicly available physical design tools remain limited to conventional two-dimensional implementations, creating a significant barrier for researchers, educators, and practitioners interested in exploring 3D integration technologies.**

The tutorial demonstrates how an existing two-dimensional (2D) OpenROAD implementation flow can be systematically extended to support both monolithic and heterogeneous multi-tier integrated circuits without requiring modifications to OpenROAD's optimization engine. Participants will learn how technology libraries, placement rows, routing resources, and design exchange formats such as LEF, DEF, and Liberty can be adapted to represent multi-tier designs while maintaining compatibility with established physical design workflows.

The session follows a complete design example that progressively evolves from a conventional synthesized RISC-V implementation into both monolithic and heterogeneous 3D realizations. Each stage of the implementation flow, including floorplanning, placement, clock-tree synthesis, power delivery network generation, routing, and physical verification, is discussed together with the practical considerations introduced by multi-tier integration. Live demonstrations and guided examples, which will be made publicly availiable, illustrate how automation scripts and open-source infrastructure enable reproducible research without relying on proprietary design tools or commercial process design kits (PDKs).

By combining theoretical foundations with hands-on implementation techniques, the tutorial equips participants with the knowledge required to develop, reproduce, and extend open-source 3D physical design flows. It also discusses current research challenges, including technology-aware placement, automated partitioning, cross-tier optimization, and heterogeneous integration, providing attendees with a foundation for future research and practical development in emerging semiconductor technologies.

***Index Terms*—3D ICs, Physical Design, OpenROAD, EDA, Heterogeneous Integration, Monolithic 3D ICs.**

Area of the Tutorial: EDA, Test and Reliability, Digital Circuits and Systems, VLSI Systems and Applications, Emerging technologies and technology trends

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I. Tutorial Description

Three-dimensional integrated circuits (3D ICs), including monolithic and heterogeneous integration technologies, have emerged as a key direction to maintaina improvements in performance, energy efficiency, and integration density beyond conventional CMOS scaling [1]. Although commercial EDA tools provide increasing support for advanced integration technologies, they remain largely proprietary and offer limited flexibility for research, education, and methodology development [2]. In contrast, existing open-source physical design frameworks, including OpenROAD, were originally developed for conventional 2D integrated circuits and do not natively support multi-tier physical design [3].

This tutorial presents a practical methodology for extending the OpenROAD physical design framework to support both monolithic and heterogeneous 3D IC implementation. Rather than providing only a conceptual overview, the tutorial follows a complete design example that is progressively transformed from a conventional 2D implementation into monolithic and heterogeneous multi-tier realizations. Participants will observe how modifications to technology libraries, process design kits (PDKs), LEF/DEF/Liberty files, placement rows, routing resources, and implementation scripts enable OpenROAD to support multi-tier placement and routing while preserving compatibility with its existing optimization flow [4].

The tutorial is organized into four modules. The first introduces the fundamentals of 3D integration technologies and discusses the limitations of conventional open-source physical design flows. The second module demonstrates the complete implementation of a monolithic 3D IC, covering floorplanning, placement, clock-tree synthesis, power delivery network generation, routing, and physical verification. The third module extends the methodology to heterogeneous 3D integration, highlighting the challenges of supporting multiple technology libraries and presenting practical solutions for heterogeneous placement using OpenROAD. Finally, participants reproduce the complete design flow through live demonstrations and guided hands-on examples, followed by a discussion of current research challenges and future directions in open-source 3D physical design.

The proposed tutorial is built upon the instructors' recent

research on open-source 3D physical design methodologies, including the development of the 3DPX framework[1] and the ongoing activities of the Horizon Europe COIN-3D project [4], [5]. The presented methodologies have been developed and validated through recent research contributions on the implementation of 3D IC, providing participants with practical insights beyond those available in existing flows. All tutorial material, including technology libraries, design examples, automation scripts, and documentation, will be distributed as an open-source repository, allowing attendees to reproduce the presented workflows and use them as a foundation for future research and education use.

A. Foundations of 3D Physical Design with OpenROAD

The tutorial begins with an introduction to the motivation for 3D integration and the limitations of conventional 2D physical design methodologies. Participants are introduced to the main 3D integration technologies, including monolithic and heterogeneous integration, together with their architectural characteristics, fabrication approaches, and application domains. The session highlights the advantages and challenges of multi-tier integration, focusing on physical design considerations rather than fabrication details.

The second part of this introductory module presents the OpenROAD physical design framework and discusses why existing open-source implementation flows cannot directly support 3D integrated circuits. The presenters introduce the common infrastructure required to enable multi-tier physical design, including adaptations to technology libraries, routing resources, placement rows, and standard design exchange formats such as LEF, DEF, and Liberty. Rather than focusing on implementation details, this section establishes the design methodology that forms the basis of both the monolithic and heterogeneous flows presented later in the tutorial.

By the end of this introductory session, participants will understand how a conventional 2D OpenROAD implementation can be systematically transformed into a flexible multi-tier physical design framework, providing the necessary foundation for the practical design examples that follow.

B. Monolithic 3D Tutorial Flow

Building upon the concepts introduced in the previous section, participants are guided through the complete implementation of a monolithic 3D integrated circuit using OpenROAD. Starting from a synthesized open-source RISC-V design, the tutorial demonstrates how a conventional 2D implementation is transformed into a two-tier monolithic design while preserving compatibility with the existing OpenROAD optimization flow.

The presenters describe each stage of the implementation process, including floorplanning, macro and standard-cell placement, clock-tree synthesis, power delivery network generation, routing, and physical verification. Throughout the demonstration, the impact of each design stage on the final implementation is discussed, highlighting the practical considerations that arise when extending a conventional 2D flow to monolithic 3D integration.

Representative implementation results are presented to illustrate the effects of monolithic integration on physical design metrics such as area utilization, routing congestion, timing, and power delivery. This section establishes the complete methodology for monolithic 3D implementation and prepares participants for the heterogeneous design flow presented in the next module.

C. 3D Heterogeneous Tutorial Flow

The third module focuses on extending OpenROAD to support heterogeneous 3D integrated circuits, where different technology libraries coexist within the same multi-tier design. Unlike conventional OpenROAD flows, heterogeneous implementations introduce additional challenges arising from different standard-cell architectures, technology nodes, row geometries, and physical design constraints. The tutorial presents these challenges and discusses the modifications required to enable heterogeneous placement within an open-source framework.

Building upon the common infrastructure introduced earlier, participants are guided through the methodology developed to adapt OpenROAD for heterogeneous multi-tier implementation. The presenters demonstrate how technology-specific libraries, placement rows, and floorplans are generated, together with the scripting infrastructure that allows the heterogeneous design to be represented internally as a conventional two-dimensional implementation. This abstraction enables OpenROAD to execute its standard placement and optimization algorithms without requiring modifications to its optimization engine, while preserving the physical correspondence between the generated 2D representation and the final heterogeneous 3D layout.

Using a complete design example, attendees follow the heterogeneous implementation flow from technology preparation and floorplanning to placement, legalization, and routing. Throughout the session, the practical challenges of heterogeneous placement are discussed, including technology-aware row generation, library scaling, cell mapping, and maintaining consistency between the virtual 2D representation and the final multi-tier design. By the end of this module, participants will understand how existing open-source physical design tools can be leveraged to implement heterogeneous 3D ICs through lightweight design-flow adaptations rather than extensive modifications to the underlying placer.

D. Hands-on Demonstration, Discussion, and Future Directions

The tutorial concludes with an interactive hands-on session in which participants reproduce the complete design flow using the provided scripts, technology libraries, and design examples. Beginning with a conventional two-dimensional implementation, attendees execute both the monolithic and heterogeneous workflows, visualize the resulting layouts, and

[1]https://github.com/coin3d-project/coin3d-project/tree/main/3DPX_testcase

analyze the generated physical design results. Throughout the demonstration, the presenters explain each stage of the flow, discuss common implementation challenges, and illustrate how the provided automation scripts simplify the transition from a standard 2D OpenROAD flow to multi-tier physical design.

The session concludes with a discussion of the presented methodologies, highlighting the practical advantages and current limitations of extending open-source physical design tools to support emerging 3D integration technologies. Particular emphasis is placed on the challenges that remain open for future research, including automated partitioning, technology-aware optimization, cross-tier timing and power optimization, and tighter integration between architectural and physical design methodologies. The tutorial ends with an open question-and-answer session, allowing participants to discuss potential applications of the presented framework in their own research and development activities.

II. Tutorial Format

The tutorial is designed as a 2-hour interactive session combining lectures with live demonstrations and guided hands-on exercises. Participants will be introduced to the theoretical foundations of monolithic and heterogeneous 3D integration before progressively extending a conventional OpenROAD physical design flow to support multi-tier implementations. The tutorial emphasizes practical experience through a complete design example, allowing attendees to reproduce the presented methodology using only open-source software.

The proposed schedule is:

- Introduction to 3D IC technologies and OpenROAD: 20 minutes
- Monolithic 3D design flow: 30 minutes
- Heterogeneous 3D design flow: 30 minutes
- Hands-on demonstration: 30 minutes
- Discussion, future directions and Q&A: 10 minutes

All tutorial material, including scripts, design examples, and documentation, will be made available to participants after the session.

III. Learning Objectives

Upon successful completion of this tutorial, participants will be able to:

LO1. Explain the principles of monolithic and heterogeneous 3D IC technologies and understand their implications for physical design.

LO2. Describe the limitations of conventional two-dimensional physical design flows and identify the key modifications required to enable multi-tier implementations.

LO3. Modify technology libraries and physical design data, including LEF, DEF, and Liberty files, to support homogeneous and heterogeneous multi-tier 3D IC designs.

LO4. Execute complete open-source physical design flows for monolithic and heterogeneous 3D ICs using the OpenROAD framework and the accompanying automation scripts.

LO5. Analyze and compare the impact of 3D integration on key physical design metrics, including area utilization, timing, routing congestion, and power delivery.

LO6. Identify current research challenges and future directions in open-source 3D physical design automation, including technology-aware optimization, automated partitioning, and cross-tier design methodologies.

IV. Target Audience

This tutorial is intended for graduate students, Ph.D. candidates, professors, academic researchers, and industry professionals working in electronic design automation (EDA), VLSI design, integrated circuits, computer architecture, and semiconductor systems. It is particularly suitable for researchers and engineers interested in emerging 3D integration technologies, open-source design automation, and next-generation physical design methodologies. The tutorial is also valuable for educators seeking practical, reproducible teaching material based on open-source EDA tools.

Participants are expected to have a basic understanding of digital integrated circuit design and standard physical design concepts, including logic synthesis, floorplanning, placement, clock-tree synthesis, and routing. Familiarity with Linux-based development environments and scripting (e.g., Tcl or Python) is beneficial but not mandatory. Prior experience with OpenROAD or 3D IC design is not required, as the tutorial begins with the fundamentals and progressively introduces the concepts, methodologies, and implementation techniques needed to build complete monolithic and heterogeneous 3D physical design flows.

V. Instructors and Credentials

A. George Rafael Goudroumanis

George Rafael Goudroumanis is a second-year Ph.D. candidate in the Department of Electrical and Computer Engineering at the University of Thessaly. Since his Master's thesis, his research has centered on physical design algorithms, spanning floorplanning, partitioning, placement, and routing. As part of a research team, he contributed to the development of a complete Place-and-Route (PnR) tool, focusing primarily on the partitioning and routing stages. Through this work, he has built a strong command of Power, Performance, and Area (PPA) metrics, together with the underlying mathematical foundations required for their implementation.

His Ph.D. research has since turned toward 3D integrated circuits, which he views as the next major frontier of the semiconductor industry and among the most promising directions for the foreseeable future. His current work aims to combine custom-developed 3D physical design algorithms with signoff methodologies – including electromigration (EM), IR drop, and thermal analysis – to enable early-stage, cross-domain analysis within the 3D-integrated ASIC design flow.

B. Maria Pantazi-Kypraiou

Maria Pantazi-Kypraiou is a Ph.D. candidate in the Department of Electrical and Computer Engineering at the University

of Thessaly. Her research initially focused on physical design automation, particularly placement algorithms and open-source implementation methodologies for integrated circuits. Through research collaborations with the University of Amsterdam, her interests expanded toward cross-layer reliability-aware system design, bridging computer architecture and physical implementation for emerging manycore and 3D integrated systems.

Her current research spans physical design, power delivery network optimization, thermal analysis, and EM reliability, with an emphasis on developing open-source methodologies for early-stage design exploration. She has contributed to the development of HotPROTON, an open-source framework for workload-aware power, thermal, IR-drop, and EM analysis, as well as methodologies that extend OpenROAD to support monolithic and heterogeneous 3D integrated circuits.

Her research interests include electronic design automation, physical design, reliability-aware computing, and 3D integrated circuits.

C. Alberto Garcia-Ortiz

Prof. Dr.-Ing. Alberto Garcia-Ortiz is Full Professor and Head of the Chair for Integrated Digital Systems (ITEM) at the University of Bremen, Germany, and Co-Director of the Institute of Electrodynamics and Microelectronics. He received his M.Sc. in Systems of Telecommunication from the Polytechnic University of Valencia (Spain) and his Dr.-Ing. in Electrical and Information Technology from the Technical University of Darmstadt. His research focuses on electronic design automation (EDA), low-power and reliable integrated circuit design, hardware architectures for AI, 3D integrated circuits, and system-level design optimization. Before joining academia, he held industrial positions at IBM Deutschland and Anafocus, contributing to microprocessor development and digital IC design.

Prof. Garcia-Ortiz is a Senior Member of IEEE and has served as General Chair, Program Chair, and Technical Program Committee Chair for numerous leading international conferences, including ISCAS, PATMOS, MOCAST, and ReCoSoC. He has received multiple prestigious awards, including the IEEE Circuits and Systems Society Industry Best Paper Award (2025), the ISLPED Best Paper Award (2022), and the EDAA Outstanding Dissertation Award. His publication record includes numerous high-impact papers in IEEE journals and conferences, with significant contributions to 3D ICs, chiplet architectures, energy-efficient computing, and dependable system design.

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